

HACKMATRIX 2K26 · ROUND 2 DOCUMENTATION

AccessLens

AI-Powered Web Accessibility Assistant — making the web understandable, actionable, and accessible, one website at a time.

PROJECT DOCUMENTATION | SUBMITTED FOR HACKMATRIX 2026

TEAM NEXORA

github.com/shreelakshmiprabhu0-ui/hackmatrix-2026-accesslens · accesslens-nine.vercel.app

Team & Problem Statement

Team Information

Team Name	NEXORA
Team Leader (Name & Contact)	Add team leader's full name and contact email / phone here before submission.
Event	HackMatrix 2026 — Round 2
Team Members	Shreelakshmi Prabhu — Frontend & Integration Shivani B — Frontend & Backend Minvitha — Backend Diya Sajin — Presentation & Documentation

Links

GitHub Repository	https://github.com/shreelakshmiprabhu0-ui/hackmatrix-2026-acesslens
Live Deployed Link	https://accesslens-nine.vercel.app/
Demo Video	Add final YouTube / Google Drive demo link before submission.

Problem Statement

Millions of websites remain inaccessible to people with visual, motor, and cognitive disabilities. Existing accessibility auditing tools are good at **detecting** technical violations, but their reports are often too technical for beginners, students, startups, and small teams to act on. Developers are left knowing something is wrong — without understanding what it means, why it matters, or how to fix it.

The core gap: Detection is not the same as understanding. AccessLens is built to bridge that gap — turning raw accessibility audit data into clear, prioritized, developer-ready action.

Project Summary & Solution

Platform Preview

Project Name	AccessLens
Platform	Web application (browser-based, no installation required)
Category	AI-powered accessibility auditing & developer assistant

Summary

AccessLens is an AI-powered accessibility assistant that scans public websites for WCAG accessibility violations and converts the results into plain-English explanations, severity-based priorities, and actionable, code-level fixes. Instead of handing developers another dense technical report, AccessLens answers the question that matters most: *"Okay — now what do I do about it?"*

Problem Being Solved

Accessibility auditing tools can flag issues like `Image elements do not have [alt] attributes`, but rarely explain what the issue means, why it matters for real users, how severe it is, or how to correctly fix it in code. This leaves beginners, students, startups, and small development teams unable to translate audit output into real improvements — so accessibility work stalls even when the data to fix it already exists.

Our Solution

- 1 Technical accessibility data is collected from an automated scan
- 2 The AccessLens engine processes and maps findings to WCAG criteria
- 3 AI generates a plain-English explanation for each issue
- 4 Issues are prioritized by severity — Critical, Moderate, Minor
- 5 Practical, code-level fixes are generated for each violation
- 6 Results are delivered on a dashboard and exportable report

Unique Selling Point & Key Features

USP — Why AccessLens Is Different

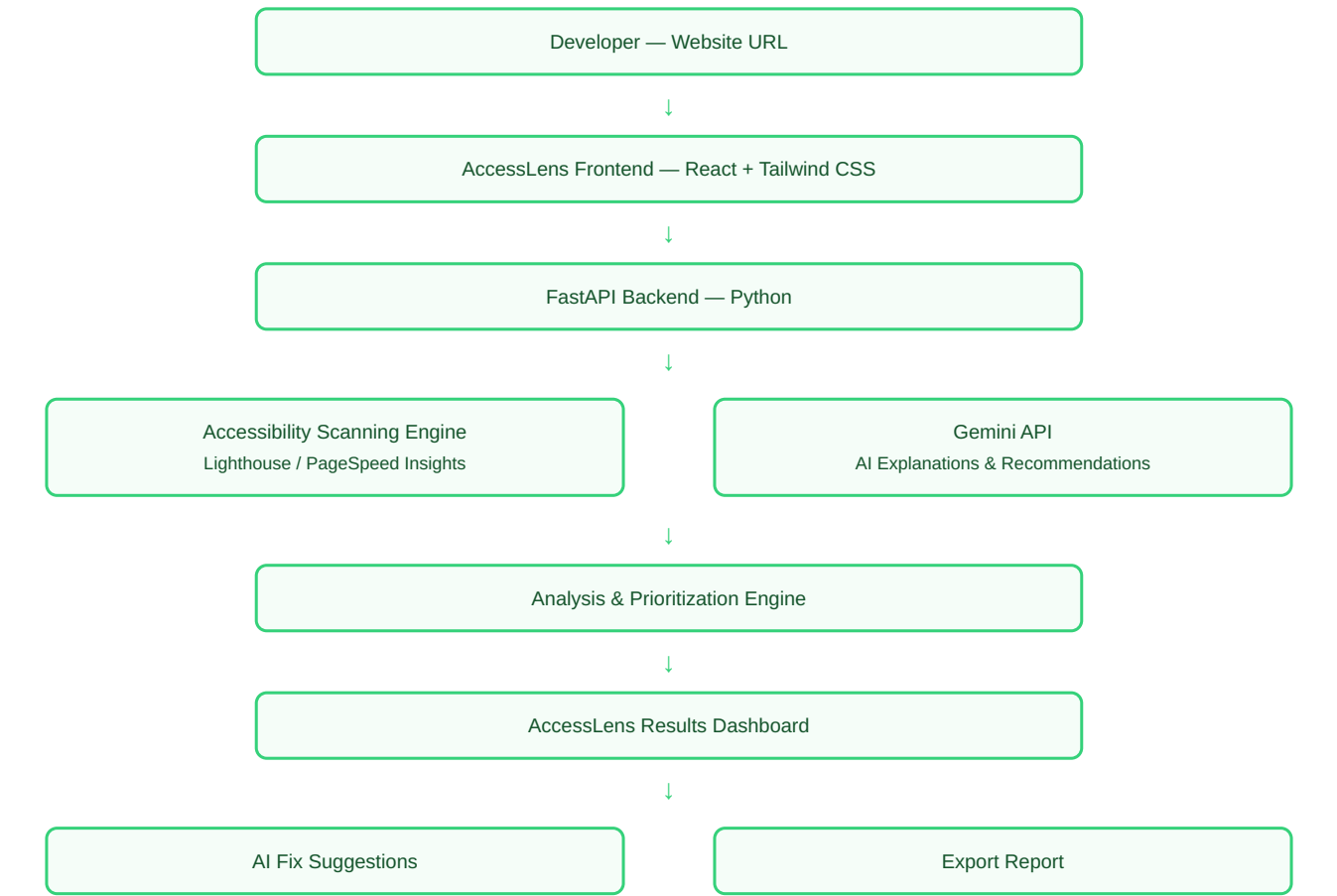
Most accessibility tools stop at **"here is what's wrong."** AccessLens goes further and helps developers reach **"here is why it matters — and here is how to fix it."** By layering an AI explanation and fix-generation engine on top of standard accessibility auditing, AccessLens functions as both a scanner *and* an accessibility learning assistant — closing the gap between detection and action rather than just adding another report to read.

- Plain-English AI explanations
- Severity prioritization
- WCAG criteria mapping
- Exportable reports
- Code-level fixes
- Learn-as-you-fix guidance

Key Features

FEATURE	DESCRIPTION
Automated Accessibility Scan	Scans any public website for accessibility issues on demand
AI-Powered Explanations	Converts technical findings into plain-English explanations
Severity Prioritization	Separates issues into Critical, Moderate, and Minor
Code-Level Fix Suggestions	Actionable, ready-to-apply recommendations for developers
Accessibility Score	Single, easy-to-understand score summarizing site health
WCAG Mapping	Connects every violation to the relevant WCAG criterion
Affected Element Details	Pinpoints exactly which elements are responsible for each issue
Accessibility Guidance	Helps developers build lasting understanding of accessibility concepts
Exportable Reports	Generates downloadable reports for documentation and review
Web-Based Interface	Fully accessible through the browser — no setup required

System Architecture



Production Flow

User → AccessLens Frontend → REST API → AccessLens Backend → (Accessibility Audit + Gemini AI) → Accessibility Analysis → Frontend Dashboard → AI Explanation, Fix Suggestions, and Export Report.

Technology Stack

Frontend

- React
- JavaScript / JSX
- HTML5 & CSS3
- Tailwind CSS
- Vite

Accessibility Engine

- Lighthouse
- Google PageSpeed Insights API
- WCAG standards & audit data

Backend

- Python
- FastAPI
- Uvicorn
- REST APIs

AI Layer

- Google Gemini API
- AI-powered explanations & guidance
- Automated fix recommendations

Deployment

Frontend	Vercel
Backend	Render
Version Control	Git & GitHub

AI Usage Disclosure

TOOL	PURPOSE
ChatGPT	Development assistance, debugging, code review, documentation
Claude	Development assistance, code review, debugging, problem-solving
Google Gemini API (in-app)	Generates plain-English explanations and fix recommendations shown to users

All AI-generated suggestions were reviewed, adapted, tested, and integrated by the team. Final architecture, implementation, and project decisions were made by Team NEXORA.

Social Impact & Future Scope

Social Impact

- Promotes digital inclusion for people with visual, motor, and cognitive disabilities
- Encourages accessible-by-default web development practices
- Makes accessibility easier for students and beginners to learn
- Reduces the time required to understand and act on accessibility issues
- Encourages WCAG-aware development across teams of every size

Future Scope

DIRECTION	DESCRIPTION
Browser Extension	Scan the current webpage directly from the browser
VS Code Extension	Detect accessibility issues live while developers write code
CI/CD Integration	Automatically scan sites during development and deployment
Enterprise Dashboard	Track accessibility across multiple websites and projects
Accessibility History	Track accessibility scores and improvement over time
AI-Assisted Remediation	Move from recommending fixes to generating safe, developer-reviewed patches

CONCLUSION

AccessLens is more than an accessibility scanner — it is an accessibility assistant designed to turn audit results into understanding, prioritization, actionable fixes, and ultimately, accessible websites.

**Accessibility should not be a checkbox.
It should be built into the web.**

Built with care by Team NEXORA

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